Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following modeling statements is most reasonable?

A. When studying the detailed bending of an elephant's trunk, it is fine to replace the trunk by a perfect sphere.
This is incorrect because trunk motion depends on shape and flexibility, so a spherical approximation removes the key geometry.

B. When tracking an elephant's location over a 50 km migration route, treating the elephant as a point mass is acceptable.
This is correct because over tens of kilometers the elephant's size is negligible compared with the travel distance.

C. You are not allowed to assume "vacuum conditions" in any situation, even if you explicitly decide to neglect drag.
This is incorrect because "ignoring air resistance" is effectively the same simplification as treating the motion as if in vacuum.

D. All of the above statements are reasonable approximations.
This is incorrect because only the large-distance point-mass approximation is justified, not the others.

Q2. Which of the following currency equations is correct?

A. 1 p = 1 £
This is incorrect because 1 penny is much smaller than 1 pound.

B. 1 = 0.01
This is incorrect because it equates dimensionless numbers without a valid reason and is not a currency conversion.

C. 1 p = 0.01 £
This is correct because 1 penny is defined as one-hundredth of a pound.

D. 100 p = 0.01 £
This is incorrect because 100 p equals 1 £, not 0.01 £.

Q3. Which option lists an incorrect SI unit for the quantity?

A. Power: watt (W)
This is correct because the watt is the SI unit of power.

B. Momentum: kilogram·meter/second (kg·m/s)
This is correct because momentum has SI units of kg·m/s.

C. Pressure: newton/meter (N/m)
This is incorrect because pressure is force per area, so it should be N/m^2 (pascal), not N/m.

D. Electric potential: joule/coulomb (J/C)
This is correct because 1 volt equals 1 joule per coulomb.

Q4. A scout drone is launched from a base at the origin and detects a target at the position vector $\vec{r} = (-2\,\hat{\imath} + 9\,\hat{\jmath})$, where distances are measured in kilounits. Which statement is correct?

A. The target is 11 kilounits from the base.
This is incorrect because the distance is $\sqrt{(-2)^2 + 9^2} = \sqrt{85} \approx 9.2$, not 11.

B. The target is 9 kilounits from the base.
This is incorrect because 9 is the y-coordinate, not the radial distance from the origin.

C. The target is 2 kilounits from the base.
This is incorrect because 2 is the magnitude of the x-coordinate, not the full distance.

D. None of the above is correct.
This is correct because the distance is $\sqrt{85} \approx 9.2$ kilounits, which is not 11, 9, or 2.

Q5. If the displacement of an object is given by $\vec{s} = \left((0.90\,\text{m/s}^3)t^3 + (3.10\,\text{m/s}^2)t^2\right)\hat{\imath}$, which statement is correct?

A. This motion must have constant acceleration.
This is incorrect because the t^3 term produces an acceleration that depends on time.

B. The magnitude of the acceleration is 3.10 m/s² for all t.
This is incorrect because differentiating twice yields an acceleration with a term proportional to t.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s².
This is incorrect because the t^2 term contributes a nonzero constant acceleration at $t = 0$.

D. The magnitude of the acceleration at $t = 0$ is 6.20 m/s².
This is correct because $a(t) = \frac{d^2s}{dt^2} = (5.40t + 6.20)\,\hat{\imath}$, so at $t = 0$ the magnitude is $6.20\,\text{m/s}^2$.

Q6. An object takes 4.0 s to move from position A ($\vec{r}_A = 1.0\,\hat{\imath} - 3.0\,\hat{\jmath}$) to position B ($\vec{r}_B = -5.0\,\hat{\imath} + 4.0\,\hat{\jmath}$) along a straight path, where length units are in meters. Which statement is correct?

A. The distance traveled is 9.2 m.
This is correct because the straight-line distance is $|\Delta\vec{r}| = \sqrt{(-6)^2 + 7^2} = \sqrt{85} \approx 9.2\,\text{m}$.

B. The speed at $t = 0$ must be 2.3 m/s.
This is incorrect because knowing only endpoints and total time does not fix the initial speed.

C. The acceleration at $t = 0$ must be 0.58 m/s².
This is incorrect because acceleration cannot be determined from just two position vectors and a travel time.

D. None of the above is correct.
This is incorrect because the distance claim in option A follows directly from the coordinate change.

Q7. A soccer ball is kicked from the ground with speed v_0 at an angle θ above the horizontal (neglect air resistance). What is the ball's speed at its highest point?

A. v_0
This is incorrect because the vertical component decreases to zero at the top, reducing the total speed.

B. $v_0 \cos\theta$
This is correct because at maximum height the vertical velocity is zero while the horizontal component remains $v_0 \cos\theta$.

C. $v_0 \sin\theta$
This is incorrect because $v_0 \sin\theta$ is the initial vertical component, which becomes zero at the top.

D. 0
This is incorrect because the ball still has horizontal velocity at the highest point.

Q8. A drone has an initial velocity $\vec{v}_0 = (15.0\,\text{m/s})\,\hat{\imath}$. Its autopilot commands a constant acceleration $\vec{a} = (-1.50\,\text{m/s}^2)\,\hat{\imath} + (2.00\,\text{m/s}^2)\,\hat{\jmath}$. Which statement is correct?

A. The x-component of the drone's velocity stays constant.
This is incorrect because a nonzero a_x changes v_x over time.

B. The y-component of the drone's velocity stays constant.
This is incorrect because a nonzero a_y changes v_y over time.

C. Both the direction of motion and the speed change.
This is correct because both velocity components vary with time when acceleration has both x and y components.

D. None of the above is correct.
This is incorrect because option C correctly describes the effect of a 2D acceleration.

Q9. In a Coulomb's-law setup, both charge magnitudes are multiplied by 5 and the separation distance is also multiplied by 5. How does the magnitude of the electrostatic force change?

A. It stays the same.
This is correct because $F \propto q_1 q_2 / r^2$ so the force scales by $25/25 = 1$.

B. It decreases by a factor of 5.
This is incorrect because the distance change affects the force by the square, not linearly.

C. It increases by a factor of 5.
This is incorrect because scaling both charges changes the numerator by 25, not 5.

D. It increases by a factor of 25.
This is incorrect because you must also include the r^2 increase (also a factor of 25) which cancels.

E. It decreases by a factor of 25.
This is incorrect because increasing the charges increases the force and cancels the distance-squared decrease, rather than compounding it.

Q10. A planet's escape speed is given by $v = \sqrt{\frac{2GM}{R}}$. Based on this relationship, which statement is correct?

A. A more massive spacecraft needs a larger escape speed from the same planet.
This is incorrect because the formula depends on the planet's M and R, not on the object's mass.

B. Decreasing the planet's mass while keeping its radius fixed increases the escape speed.
This is incorrect because smaller M makes v smaller, not larger.

C. Increasing the planet's radius while keeping its mass fixed increases the escape speed.
This is incorrect because larger R appears in the denominator and therefore lowers v.

D. A planet with larger mass and smaller radius has a higher escape speed.
This is correct because v increases with M and decreases with R according to $v = \sqrt{2GM/R}$.

Q11. A force $\vec{F} = (2.50\,\text{N})\,\hat{\imath} + (6.00\,\text{N})\,\hat{\jmath}$ acts on a particle as it moves along the x axis from $x = 1.20\,\text{m}$ to $x = 4.80\,\text{m}$. What is the work (in joules) done by the force?

A. 9.00 J
This is correct because $W = \vec{F} \cdot \Delta\vec{r} = F_x \Delta x = (2.50)(3.60) = 9.00\,\text{J}$ since the motion is purely along x.

B. 21.6 J
This is incorrect because it wrongly includes the y-component even though there is no y displacement.

C. 3.60 J
This is incorrect because it confuses displacement ($\Delta x = 3.60$ m) with work, missing the force factor.

D. 16.0 J
This is incorrect because it comes from an arithmetic or component-selection error in the dot product.

Q12. A 3.0 kg cart moving at 4.0 m/s hits a rigid barrier and rebounds at 2.0 m/s in the opposite direction. The collision lasts 0.15 s. What is the magnitude of the average force on the cart during the collision?

A. 120 N
This is correct because $|F_{\text{avg}}| = |\Delta p|/\Delta t = |3.0(-2.0 - 4.0)|/0.15 = 120\,\text{N}$.

B. 40 N
This is incorrect because it effectively uses only part of the velocity change (or uses the wrong final sign).

C. 80 N
This is incorrect because it comes from dividing by an incorrect collision time or using an incorrect momentum change.

D. None of the above is correct.
This is incorrect because the impulse calculation gives a magnitude of 120 N.

Q13. A 62 kg goalkeeper is initially at rest and catches a 145 g ball moving at 110 km/hr. Assuming the ball sticks in the glove, what is the goalkeeper's speed immediately after the catch?

A. 0.071 m/s
This is correct because momentum conservation for a perfectly inelastic catch gives $v = \frac{(0.145)(110\,\text{km/hr} \times 1000/3600)}{62.145} \approx 0.071\,\text{m/s}$.

B. 0.093 m/s
This is incorrect because it typically results from not converting km/hr to m/s correctly or using the wrong total mass.

C. 30.6 m/s
This is incorrect because it confuses the ball's incoming speed with the combined post-catch speed.

D. 2.1 m/s
This is incorrect because it overestimates the result by neglecting the large goalkeeper mass in the denominator.
